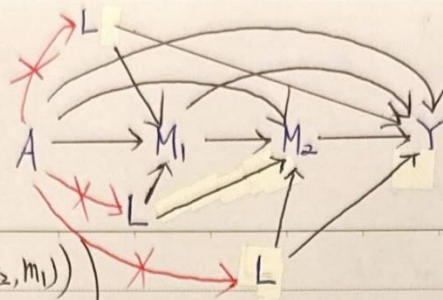


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$$\left(\begin{array}{l} Y(a, m_1, m_2) \perp\!\!\!\perp (M_1(e_1), M_2(e_2, m_1)) \\ M_2(e_2, m_1) \perp\!\!\!\perp M_1(e_1) \end{array} \right)$$

$$= \sum_{m_1} \sum_{m_2} E[Y(a, m_1, m_2)] P(M_2(e_2, m_1) = m_2) P(M_1(e_1) = m_1)$$

$$\left(\begin{array}{l} Y(a, m_1, m_2) \perp\!\!\!\perp A \\ Y(a, m_1, m_2) \perp\!\!\!\perp M_1 \mid A \\ Y(a, m_1, m_2) \perp\!\!\!\perp M_2 \mid A, M_1 \end{array} \right)$$

$$= \sum_{m_1} \sum_{m_2} E[Y(a, m_1, m_2) \mid M_2 = m_2, M_1 = m_1, A = a] P(M_2(e_2, m_1) = m_2) \times P(M_1(e_1) = m_1)$$

$$\left(\begin{array}{l} \text{Consistency} \\ M_1(e_1) \perp\!\!\!\perp A \\ M_2(e_2, m_1) \perp\!\!\!\perp A \\ M_2(e_2, m_1) \perp\!\!\!\perp M_1 \mid A \end{array} \right)$$

$$= \sum_{m_1} \left(\sum_{m_2} E(Y \mid M_2 = m_2, M_1 = m_1, A = a) P(M_2 = m_2 \mid M_1 = m_1, A = e_2) \right) \times P(M_1 = m_1 \mid A = e_1)$$

Estimation

$$E(Y \mid A, M_1, M_2) = \theta_0 + \theta_1 M_1 + \theta_2 M_2 + \theta_3 A$$

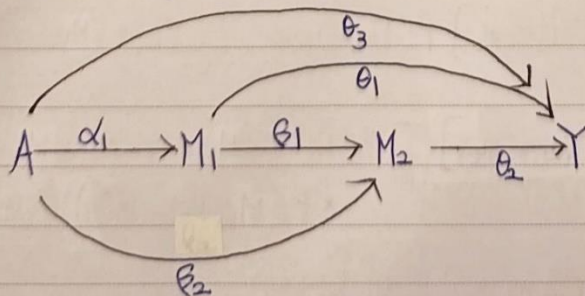
$$E(M_2 \mid A, M_1) = \beta_0 + \beta_1 M_1 + \beta_2 A$$

$$E(M_1 \mid A) = \alpha_0 + \alpha_1 A$$

$$\Rightarrow PSE_0 = \theta_1$$

$$PSE_2 = \theta_2 \beta_2$$

$$PSE_{12} + PSE_1 = \alpha_1 \beta_1 \theta_2 + \alpha_1 \theta_1$$



★ Complete Decomposition

Define $\phi_c(a_1, a_2, a_3, a_4) = E[Y(a_1, M_1(a_2), M_2(a_3, M_1(a_4)))]$

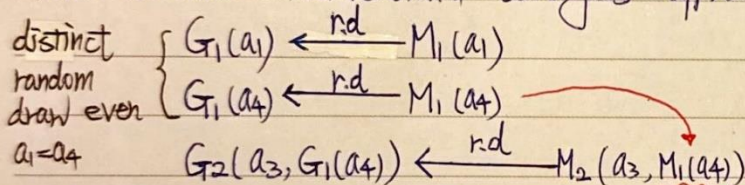
$$TE = [\phi_c(1, 1, 1, 1) - \phi_c(1, 1, 1, 0)] \Leftrightarrow PSE_{12}$$

$$+ [\phi_c(1, 1, 1, 0) - \phi_c(1, 1, 0, 0)] \Leftrightarrow PSE_2$$

$$+ [\phi_c(1, 1, 0, 0) - \phi_c(1, 0, 0, 0)] \Leftrightarrow PSE_1$$

$$+ [\phi_c(1, 0, 0, 0) - \phi_c(0, 0, 0, 0)] \Leftrightarrow PSE_0$$

- It is generally unidentifiable through empirical data if no stronger assumptions can be used.
- In order to assess the effects under complete decomposition, we consider interventional analogues approach.



Define $\psi_c(a_1, a_2, a_3, a_4) = E[Y(a_1, G_1(a_2), G_2(a_3, G_1(a_4)))]$

then

$$ITE = [\psi_c(1, 1, 1, 1) - \psi_c(1, 1, 1, 0)] \Leftrightarrow iPSE_{12} \quad (ITE \neq TE)$$

$$+ [\psi_c(1, 1, 1, 0) - \psi_c(1, 1, 0, 0)] \Leftrightarrow iPSE_2$$

$$+ [\psi_c(1, 1, 0, 0) - \psi_c(1, 0, 0, 0)] \Leftrightarrow iPSE_1$$

$$+ [\psi_c(1, 0, 0, 0) - \psi_c(0, 0, 0, 0)] \Leftrightarrow iPSE_0$$

The identification of $\psi_c(a_1, a_2, a_3, a_4)$

$$\psi_c(a_1, a_2, a_3, a_4) = \sum_{m_1} \sum_{m_2} E[Y(a_1, m_1, m_2) | G_1(a_2) = m_1, G_2(a_3, G_1(a_4)) = m_2] P(G_1(a_2) = m_1) \times P(G_2(a_3, G_1(a_4)) = m_2)$$

$$= \sum_{m_1} \sum_{m_2} E[Y(a_1, m_1, m_2)] P(M_1(a_2) = m_1) P(M_2(a_3, G_1(a_4)) = m_2)$$

$$= \sum_{m_1} \sum_{m_2} E[Y(a_1, m_1, m_2)] \left(\sum_{m_1^*} P(M_2(a_3, G_1(a_4)) = m_2 | G_1(a_4) = m_1^*) \times P(G_1(a_4) = m_1^*) \right) P(M_1(a_2) = m_1)$$

$$= \sum_{m_2} \left(\sum_{m_1} E[Y(a_1, m_1, m_2)] P(M_1(a_2) = m_1) \right)$$

$$\times \left(\sum_{m_1^*} P(M_2(a_3, m_1^*) = m_2) P(M_1(a_4) = m_1^*) \right)$$

$$\left(\begin{array}{l} Y(a_1, m_1, m_2) \perp\!\!\!\perp A \\ Y(a_1, m_1, m_2) \perp\!\!\!\perp M_1 \mid A \\ Y(a_1, m_1, m_2) \perp\!\!\!\perp M_2 \mid A, M_1 \\ M_2(a_3, m_1^*) \perp\!\!\!\perp A \\ M_2(a_3, m_1^*) \perp\!\!\!\perp M_1 \mid A \\ M_1(a_2) \perp\!\!\!\perp A \end{array} \right)$$

$$= \sum_{m_2} \left(\sum_{m_1} E[Y \mid M_2 = m_2, M_1 = m_1, A = a_1] P(M_1 = m_1 \mid A = a_2) \right) \left(\sum_{m_1^*} P(M_2 = m_2 \mid M_1 = m_1^*, A = a_3) P(M_1 = m_1^* \mid A = a_4) \right)$$

★ Relationship among strategies

Essentially, $\phi_{TW}, \phi_F, \phi_B, \phi_C$ follows hierarchical relationship

$$\phi_{TW}(a, e) = \phi_F(a, e, e) = \phi_B(a, e, e) = \phi_C(a, e, e, e)$$

$$\phi_F(a, e_1, e_2) = \phi_C(a, e_1, e_2, e_1)$$

$$\phi_B(a, e_1, e_2) = \phi_C(a, e_1, e_2, e_2)$$

Although ϕ_C can provide detailed insight into the mechanism, more assumptions are necessary for ϕ_C

how details of mechanism



how many assumptions

Estimation

$$E(Y \mid M_2, M_1, A) = \theta_0 + \theta_1 M_1 + \theta_2 M_2 + \theta_3 A$$

$$E(M_2 \mid M_1, A) = \beta_0 + \beta_1 M_1 + \beta_2 A$$

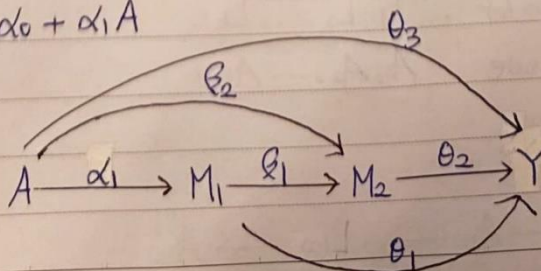
$$E(M_1 \mid A) = \alpha_0 + \alpha_1 A$$

$$IPSE_0 = \theta_3$$

$$IPSE_1 = \alpha_1 \theta_1$$

$$IPSE_2 = \beta_2 \theta_2$$

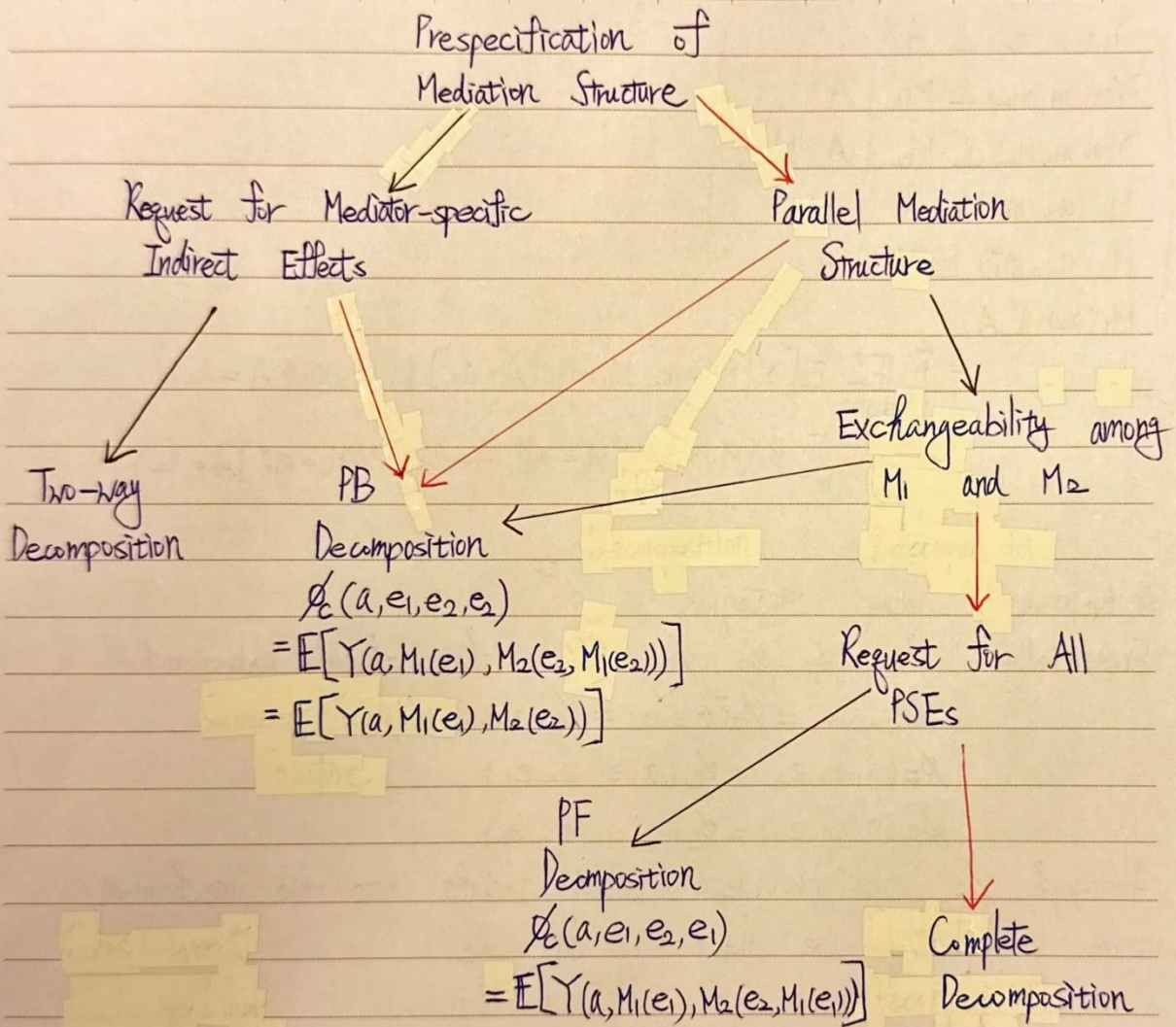
$$IPSE_{12} = \alpha_1 \beta_1 \theta_2$$



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Time-varying

Panel data $\subseteq$ Longitudinal data

repeated measurement $\rightarrow$ varying time point

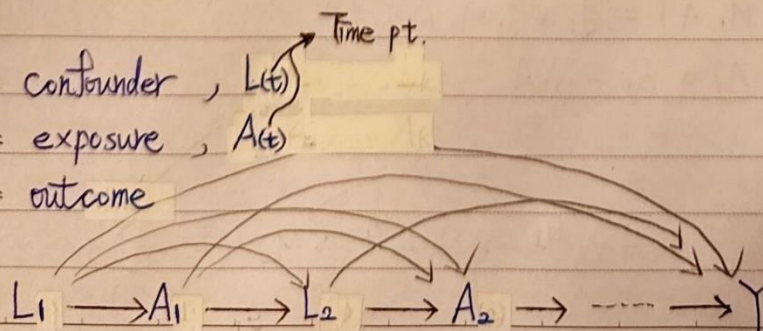
Repeated measurement

Ex.

L : confounder, $L(t)$

A : exposure, $A(t)$

Y : outcome



Difference-in-Difference (DiD)

DiD is a method for estimating the effect of

- { policy interventions (before-after comparison)
- { exogenous shock (underlying time trend)

when randomized experiments are not feasible. Cross-Sectional comparison

★ The key idea behind DiD is to compare changes in outcomes over time between treated and control groups. Time-series comparison

Why DiD?

- i) Intuitive visuals
- ii) Light data demand
- iii) Extendable
- iv) Transparent

DiD cleverly harnesses inter-temporal variation between groups in two complementary ways to fight omitted variable bias

After (T=1) Before (T=0)

Treated (A=1) $E[Y_{1(1)} | A=1]$ $E[Y_{0(0)} | A=1]$ ★ We can't observe

Control (A=0) $E[Y_{1(0)} | A=0]$ $E[Y_{0(0)} | A=0]$ $E[Y_{0(1)} | A=1]$

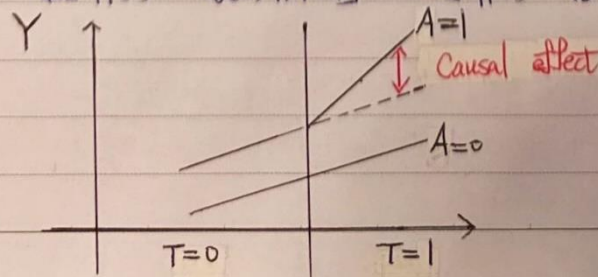
$Y_{t(a)}$ is potential outcome under treatment $A=a$ at time $T=t$ ⇕

★ DiD main interest in

$$E[Y_{1(1)} - Y_{1(0)} | A=1]$$

i) Parallel trends assumption

$$E[Y_{1(0)} - Y_{0(0)} | A=1] = E[Y_{1(0)} - Y_{0(0)} | A=0]$$



ii) fact = no causal effect of A on Y_0

$$Y_0(1) = Y_0(0) = Y_0$$

★ Identification

$$\begin{aligned} & E[Y_1(1) - Y_1(0) | A=1] \\ &= E[Y_1(1) | A=1] - E[Y_1(0) | A=1] \\ &= E[Y_1(1) - Y_0(1) | A=1] - E[Y_1(0) - Y_0(0) | A=1] \quad (\text{no causal effect of } A \text{ on } Y_0) \\ &= E[Y_1(1) - Y_0(1) | A=1] - E[Y_1(0) - Y_0(0) | A=0] \quad (\text{parallel trend assumption}) \\ &= E[Y_1 - Y_0 | A=1] - E[Y_1 - Y_0 | A=0] \quad (\text{consistency}) \end{aligned}$$

★ Estimation

DiD estimator via linear model

$$E(Y_T | A) = \beta_0 + \beta_1 T + \beta_2 A + \theta(A \times T)$$

where

$$\begin{aligned} \text{DiD} &\triangleq E[Y_1 - Y_0 | A=1] - E[Y_1 - Y_0 | A=0] \\ &= (\beta_1 + \beta_2 + \theta) - \beta_2 - \beta_1 \\ &= \theta \end{aligned}$$

Thus DiD matches the interaction term coef.

	Treated ($A=1$)	Not Treated ($A=0$)	Difference over treatment
Pre ($T=0$)	$\beta_0 + \beta_2$	β_0	β_2
Post ($T=1$)	$\beta_0 + \beta_1 + \beta_2 + \theta$	$\beta_0 + \beta_1$	$\beta_2 + \theta$
Change Over Time	$\beta_1 + \theta$	β_1	θ

↑ capture the treatment effect + general time trends
↑ capture only general time trends
↑ Diff-in-Diff

Time-varying treatment

Consider a time-varying dichotomous treatment

$$A(t) = \begin{cases} 1 & \text{if receive treatment at time } t, t=0,1,\dots,T \\ 0 & \text{otherwise} \end{cases}$$

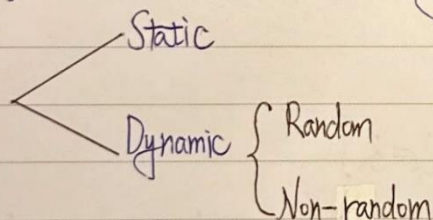
The treatment hist through time t is

$$\underline{A}(t) = (A(0), \dots, A(t))$$

and denote entire treatment hist by

$$\underline{A} = \underline{A}(T) = (A(0), \dots, A(T))$$

Define the treatment regimes (plans, strategies)



★ Static regimes

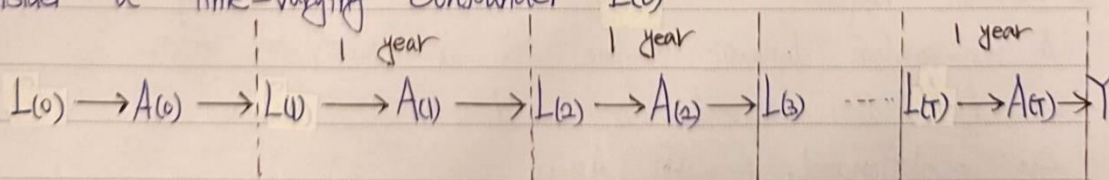
$$\underline{a} = (a(0), \dots, a(T))$$

(Ex. Always Treat $\underline{a} = (1, \dots, 1) = \underline{1}$
 Never Treat $\underline{a} = (0, \dots, 0) = \underline{0}$
 Treatment every other period $\underline{a} = (1, 0, 1, 0, \dots)$)

there are many possible causal effects for $\underline{a}$
 each of them defined by contrast of $Y(\underline{a})$ and $Y(\underline{a}^*)$

$$E[Y(\underline{a})] - E[Y(\underline{a}^*)]$$

Consider a time-varying confounder $L(t)$



Define $\phi(\underline{a}) = E[Y(\underline{a})]$

• Identification of $\phi(\underline{a})$

$$\phi(\underline{a}) = \int_{\underline{l}(0)} E[Y(\underline{a}) | L(0) = \underline{l}(0)] P(L(0) = \underline{l}(0))$$

$$\begin{aligned}
 (Y(a) \perp\!\!\!\perp A(0) \mid L(0)) &= \int_{L(0)} E[Y(a) \mid L(0)=l(0), A(0)=a(0)] P(L(0)=l(0)) \\
 &= \int_{L(0)} \left\{ \sum_{L(1)} E[Y(a) \mid L(0)=l(0), L(1)=l(1), A(0)=a(0)] \right. \\
 &\quad \left. P(L(1)=l(1) \mid L(0)=l(0), A(0)=a(0)) \right\} \\
 &\quad P(L(0)=l(0))
 \end{aligned}$$

$$\begin{aligned}
 &\star \text{ Sequential Exchangeability} \\
 &Y(a) \perp\!\!\!\perp A(1) \mid A(0), L(0), L(1) \\
 &Y(a) \perp\!\!\!\perp A(2) \mid A(0), A(1), L(0), L(1), L(2) \\
 &\vdots \\
 &Y(a) \perp\!\!\!\perp A(t) \mid A(t-1), L(t)
 \end{aligned}$$

$$\begin{aligned}
 &= \int_{\underline{L}} E[Y(a) \mid A=a, \underline{L}=\underline{L}] \\
 &\quad \prod_{t=0}^{\infty} P(L(t)=l(t) \mid \underline{L}(t-1)=\underline{L}(t-1), A(t-1)=a(t-1))
 \end{aligned}$$

$$\begin{aligned}
 (\text{Consistency } Y(a)=Y \text{ if } A=a) \\
 &= \int_{\underline{L}} E(Y \mid A=a, \underline{L}=\underline{L}) \\
 &\quad \prod_{t=0}^{\infty} P(L(t)=l(t) \mid \underline{L}(t-1)=\underline{L}(t-1), A(t-1)=a(t-1))
 \end{aligned}$$

If $\underline{L}$ is conti

$$\phi(a) = \int_{\underline{L}} E(Y \mid A=a, \underline{L}=\underline{L}) \prod_{t=0}^{\infty} f(l(t) \mid \underline{L}(t-1), a(t-1)) d\underline{L}$$

$\star$ Dynamic regimes

A non-random dynamic regime is a treatment strategy in which $a(t)$ depends in a deterministic manner on the $\underline{L}(t)$ and possibly $\underline{L}(t-1)$.

(Ex.

Take the treatment at time t iff the cell count > 100 for 3 consecutive weeks and the patient was not on treatment at week $t-1$

- A non-random dynamic g is a collection of

$$\{g_t(a_{t-1}, l(t)), t=0, \dots, T\}$$

map
↓
 $\{0, 1\}$

Ex.

$$g_t(a_{t-1}, l(t)) = \begin{cases} 1 & \text{if } a_{t-1}=0 \text{ and } l(t), l(t-1), l(t-2) > 1000 \\ 0 & \end{cases}$$

- A random dynamic regime is a treatment strategy where $a(t)$ depends in a probabilistic way on $l(t)$ and possibly $a(t-1)$

Ex.

if the cell counts has been > 100 for 3 consecutive weeks, randomize the subject to take treatment at week t w/

$$\begin{cases} 0.8 & \text{take} \\ 0.2 & \text{not take} \end{cases}$$

randomization probability

- Let g be regime and let $Y(g)$ denote the counterfactual outcome had regime g been followed.
- If high value of Y are consider beneficial, then the optimal regime g_{opt} is

$$g_{opt} = \arg \min_g E[Y(g)]$$

- In fact, we need only try to find optimal regime among the deterministic regime as no random strategy can ever preferred to the optimal deterministic strategy.

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(Remark.

Random regime remain scientifically necessary because, before the trial it is unknown which deterministic regime is optimal.

★ We need strengthened identifiability assumptions

1. Consistency.

for any g , we have $A_t = g_t(A_{t-1}, L_t)$

then $Y(g) = Y$ and $L_g = L$

↑ counterfactual under g

2. Sequential Exchangeability

$Y(g) \perp\!\!\!\perp A_t \mid L_t, \{A(s) = g_s(A(s-1), L(s)), s=0, \dots, t\}$